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ORIGINAL ARTICLE

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Identification of preterm birth in women with threatened preterm labour between 34 and 37 weeks of gestation

Cenk Gezer, Atalay Ekin, Ulas Solmaz, Alkim Gulsah Sahingoz Yildirim, Askin Dogan and Mehmet Ozeren

Department of Obstetrics and Gynecology, Tepecik Training and Research Hospital, Izmir, Turkey

ABSTRACT

The aim of this study was to assess whether serum markers would be useful as a new predictor of preterm birth in patients with spontaneous, late preterm labour. Patients diagnosed with late preterm labour were divided into preterm delivery (229 patients) and term delivery (178 patients) groups. The two groups were compared in terms of clinical characteristics and levels of serum markers (leukocyte subtypes, platelet, C-reactive protein [CRP], neutrophil to lymphocyte ratio [NLR] and platelet to lymphocyte ratio [PLR]), which were obtained at admission. The levels of leukocyte ($p < .001$), neutrophil ($p < .001$), CRP ($p = .001$), NLR ($p < .001$) and PLR ($p = .003$) were significantly higher, whereas lymphocytes ($p = .012$) were significantly lower in the preterm delivery group, compared to the term delivery group. On multivariate regression analysis, NLR positive was the most powerful predictive variable (OR = 1.41; 95%CI: 1.32–1.51; $p = .005$). NLR had the highest area under curve (0.711; 95%CI 0.662–0.760) in predicting preterm birth and a NLR >6.2 had the highest sensitivity (65.1%) and specificity (62.5%). High NLR at admission is an independent predictor of preterm birth in patients with spontaneous, late preterm labour.

IMPACT STATEMENT

- **What is already known on this subject:** Preterm birth accounts for 5–12% of all births, and is a major factor associated with perinatal morbidity and mortality worldwide. However, more than 70% of preterm births occur at late preterm between $34^{0/7}$ and $36^{6/7}$ weeks of gestation. The central role of systemic and subclinical infections in preterm labour is well documented. Intrauterine infection leading to delivery can be measured by using a variety of laboratory parameters.
- **What do the results of this study add:** Neutrophil to lymphocyte ratio is an inexpensive, easily interpretable and promising haematologic parameter that is widely available. This study explored the association of high neutrophil to lymphocyte ratio with the risk of preterm birth in women with preterm labour between 34 and 37 weeks of gestation.
- **What are the implications of these findings for clinical practice and/or further research:** Neutrophil to lymphocyte ratio could be used in combination with existing markers to improve detection rates of preterm birth. Concomitant use of markers could be more powerful than measuring any of the individual markers alone.

KEYWORDS
Lymphocyte; neutrophil; preterm delivery; preterm labour

Introduction

Preterm birth accounts for 5–12% of all births, and is a major factor associated with perinatal morbidity and mortality worldwide (Joseph et al. 2002). Most of the predictive strategies of preterm birth are targeted towards the births before 32 weeks of gestation, due to high mortality (Slattery and Morrison 2002; Raju 2006). However, more than 70% of preterm births occur at late preterm between $34^{0/7}$ and $36^{6/7}$ weeks of gestation (Goldenberg et al. 2008). In addition, neonates born late preterm also had a significantly increased risk of adverse outcomes, compared with those born at term (Engle et al. 2007).

Approximately 70–80% of preterm births are the result of spontaneous onset of labour (Slattery and Morrison 2002). However, nearly 50% of the women hospitalised for preterm labour deliver at term (McPheeters et al. 2005). In order to avoid unnecessary interventions, mechanisms leading to premature birth in patients with preterm labour should be investigated. Although what initiates preterm labour has not been clarified fully, the central role of systemic and subclinical infections are well documented in several studies (Goldenberg et al. 2000; Gotsch et al. 2009). Currently, transvaginal ultrasound measurement of cervical length and cervicovaginal foetal fibronectin testing are reliable prognostic methods for women with spontaneous labour before 34 weeks (Peaceman et al. 1997; American College of Obstetricians and Gynecologists 2001; Sotiriadis et al. 2010). Therefore, there is a need for additional markers predicting preterm birth among patients with spontaneous labour at late preterm.

Intrauterine infection leading to delivery can be measured by using a variety of laboratory parameters.

CONTACT Cenk Gezer drcenkgezer@gmail.com Department of Perinatology, Tepecik Training and Research Hospital, Gaziler St, No: 468, 35120 Izmir, Turkey

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Inflammatory markers such as cytokines and C-reactive protein (CRP) were found to be at high concentrations in serum of women with symptoms of preterm labour who later had a preterm delivery (Goldenberg et al. 2000). In recent years, elevated levels of neutrophil to lymphocyte ratio (NLR) and platelet to lymphocyte ratio (PLR) have been proposed as diagnostic and prognostic factors in various diseases (Cho et al. 2009; Kaya et al. 2014; Kemal et al. 2014). Moreover, Kim et al. (2011) performed an encouraging study including women with spontaneous labour between 22 and 37 gestational weeks. Their study revealed that NLR was significantly higher in preterm delivery group than term delivery group and was able to discriminate between preterm and term delivery (Kim et al. 2011).

In this study, we aimed to evaluate whether changes in the values of leukocyte subtypes, platelet, CRP, NLR and PLR could be used as a novel predictive marker of preterm birth in patients with spontaneous, late preterm labour.

Materials and methods

This study was retrospectively conducted between January 2010 and January 2015 in a tertiary care centre, which is the most widely circulating hospital in its region. Institutional ethics committee approval was obtained from our centre and the study was performed in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki. The database of our perinatology department was used to identify pregnancies with spontaneous, preterm labour. Spontaneous preterm labour was defined as presence of four or more regular uterine contractions within a 30-minute period, each lasting at least 30 seconds and cervical change (dilatation of $\leq$3 cm) with intact membranes. Initially, the obstetric records of all patients with a diagnosis code of preterm labour were examined. When more than one admission was found for preterm labour, the gestational week of the first admission was recorded for analysis. Although not all patients with preterm labour delivered within first hospitalisation, the birth records of these pregnancies were also collected. Since our primary outcome of interest was the interval between onset of labour and delivery, pregnancies lost to follow-up were excluded.

Women with spontaneous late preterm labour between $34^{0/7}$ and $36^{6/7}$ weeks singleton gestations were selected for the study. Computer based patient records were then thoroughly reviewed for the following information: age, gravidity, parity, body mass index, type of conception (spontaneous versus *in vitro* fertilisation), smoking habit, foetal gender, history of preterm birth, antenatal bleeding, gestational age at admission and delivery, delivery interval, type of delivery (spontaneous versus indicated and vaginal versus Caesarean) and maternal serum markers (CRP level and counts of platelet and leukocyte subtypes) upon admission. The NLR was defined as the absolute neutrophil count divided by the absolute lymphocyte count, and the PLR was defined as the absolute platelet count divided by the absolute lymphocyte count.

According to our protocol, all patients with symptoms of preterm labour between 34 and 37 weeks of gestation were hospitalised for surveillance after diagnosis. Upon admission, a speculum was introduced into the vagina and unless rupture of membranes was confirmed, the cervix was examined digitally for dilatation and effacement. Ultrasonography was performed and full blood count, as well as CRP level, were obtained from all patients. Additionally, antenatal corticosteroid and tocolytic therapies were not administered beyond 34 weeks in our centre. In the absence of foetal or maternal indication for delivery, conservative management is recommended. Foetal heart rate and uterine contractions were monitored until uterine activity and cervical changes had been stopped. If preterm labour did not progress to delivery, all patients were discharged.

Gestational age was calculated on the basis of the last menstrual period and confirmed by ultrasonographic measurements in the first or second trimester. Exclusion criteria were women with multiple gestations, cervical dilatation >3 cm at presentation, cervical cerclage, major uterine anomaly, acute or chronic inflammatory diseases, systemic diseases, history of prior preterm birth, short cervix at mid trimester ultrasound (<25 mm), pregnancies complicated by placenta previa, oligohydramnios, polyhydramnios, stillbirth, major foetal or chromosomal anomalies and those who underwent indicated delivery before $37^{0/7}$ weeks of gestation. Short cervix at midtrimester was defined as the transvaginal measurement of cervical length between 16 and 24 weeks, according to a standardised technique (To et al. 2001). Patients who had significant obstetric or medical complications mandating early delivery, such as non-reassuring foetal status, clinical chorioamnionitis, placental abruption, significant antepartum haemorrhage and intrauterine growth restriction (estimated birth weight is < 10th percentile for gestational age) were also excluded from the study. Clinical chorioamnionitis was defined as presence of maternal fever ($\geq$38 °C) with two of the following signs in the absence of other potential infectious sources: maternal tachycardia (>100 beats/min), maternal leukocytosis (15000 cells/mm^3), uterine tenderness, foetal tachycardia (>160 beats/min) and foul odour of amniotic fluid.

Patients with preterm labour were divided into two groups based on time elapsed between onset of labour and delivery. Patients in group one delivered before 37 weeks (preterm delivery group), whereas those in group two delivered at or after 37 weeks (term delivery group). The two groups were compared in terms of clinical characteristics and levels of serum markers which were obtained at admission. CRP levels were measured by immunotubidimetric method (Abbott Aeroset 1600 Autoanalyser; Abbott Reagents, Wiesbaden, Germany). Subtypes of leukocyte and platelet counts were analysed as part of the automated complete blood count by flow cytometry with EDTA plasma (Coulter LH 750 analyser; Beckman Coulter, Galway, Ireland).

Statistical analyses were performed using SPSS (SPSS for Windows version 20.0, SPSS Inc., Chicago, IL). Our sample size yields more than 80% power to detect a 0.5 SD difference in levels of serum markers between groups with a significance level of alpha = 0.05. Continuous data were expressed as mean ± standard deviation, while categorical data were presented as number of patients and percentages.

Normal distribution pattern of continuous variables was tested by Kolmogorov–Smirnov method. Student's *t*-test was used to analyse normally distributed variables and the chi-square test was used for categorical variables. To determine which serum markers predict the preterm delivery, univariate analysis and subsequently multivariate regression analysis were performed. A receiver operating characteristic (ROC) curve was used to evaluate sensitivity, specificity and optimal cut-off values of each marker in predicting preterm delivery. A p value $<.05$ was considered to be statistically significant for all analyses. Relationship between NLR and time from onset of labour to delivery was analysed with Cox-proportional hazard model. The log-rank test was utilised to assess the statistical difference between survival curves.

Results

During the 5-year period, a total of 817 patients were identified as having spontaneous, preterm labour. Of these pregnancies, 476 were diagnosed between $34^{0/7}$ and $36^{6/7}$ weeks of gestation. Sixty-nine patients who did not meet the inclusion criteria were excluded from the study. Eventually, the study population included 407 patients with spontaneous, late preterm labour; of these, 229 (56.3%) patients delivered before 37 weeks of gestation and 178 (43.7%) patients delivered at term. The clinical characteristics of the subjects are presented in Table 1. The mean time from onset of labour to delivery (3.1 versus 26.4 days, $p<.001$) and gestational age at delivery (35.6 versus 39.2 weeks, $p<.001$) were significantly shorter in the preterm delivery group compared with those in the term delivery group. As expected, foetuses in the preterm delivery group had significantly lower birthweight (2625 versus 3430 g, $p<.001$).

Levels of leukocyte subtypes, platelet, CRP, NLR and PLR were compared between the two groups (Table 2). The mean values of leukocyte ($p<.001$), neutrophil ($p<.001$), CRP ($p=.001$), NLR ($p<.001$) and PLR ($p=.003$) were significantly higher, whereas lymphocytes ($p=.012$) were significantly lower in preterm delivery group compared to term delivery group. Univariate analysis indicated that neutrophil count ($p=.042$), lymphocyte count ($p=.01$), CRP level ($p=.01$) and NLR ($p=.001$) were significant predictors of preterm delivery (Table 3). However, NLR was remained as the most powerful predictive variable in both univariate (Odds ratio [OR] = 1.56; 95%CI: 1.29–1.78; $p=.001$) and multivariate regression analyses (OR = 1.42; 95%CI: 1.32–1.51; $p=.005$) (Table 3).

Furthermore, ROC curve analysis suggested that NLR had the highest area under curve (AUC = 0.711; 95% confidence interval (CI) 0.662–0.760; $p<.001$) in predicting spontaneous, preterm delivery (Table 4). The optimum NLR cut-off point for preterm delivery was 6.2, with a sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV) of 65.1%, 62.5%, 69.3% and 57.9%, respectively. The same analysis for other markers is summarised in Table 4. The Cox proportional hazard model revealed that patients with NLR positive (>6.2) had a significantly shorter interval between onset of labour and delivery than patients with NLR negative (≤ 6.2) (mean 8.6 versus 20.2 days; log rank, $p<.001$) (Figure 1).

Table 1. Clinical characteristics of study population[a].

	Preterm delivery ($n=229$)	Term delivery ($n=178$)	p value
Maternal age (years)	25.9 ± 5.1	26.8 ± 5.3	.110
BMI (kg/m^2)	26.4 ± 2.6	25.8 ± 2.9	.168
Nulliparity	101 (44.1)	63 (35.4)	.075
Parity	0.9 ± 1.06	1.02 ± 1	.220
Abortion	0.42 ± 0.78	0.39 ± 0.73	.789
Conception by IVF	8 (3.5)	5 (2.8)	.485
Foetal gender (F/M)	110/119 (0.92)	92/86 (1.07)	.466
Smoking	13 (5.7)	8 (4.5)	.590
Gestational age at admission (weeks)	35.2 ± 0.9	35.4 ± 0.9	.231
34	107 (46.7)	75 (42.1)	
35	66 (28.8)	50 (28.1)	
36	56 (24.5)	53 (29.8)	
Gestational age at delivery (weeks)	35.6 ± 0.9	39.2 ± 1.1	<.001
34	69 (30.1)		
35	57 (24.9)		
36	103 (45)		
37		14 (7.9)	
38		33 (18.5)	
39		55 (30.9)	
40		58 (32.6)	
41		18 (10.1)	
Onset of labour to delivery interval (days)	3.1 ± 2.2	26.4 ± 10.6	<.001
Caesarean delivery	40 (17.5%)	26 (14.6%)	.438
Birthweight	2625 ± 244.9	3430 ± 268.9	<.001

[a]Data are presented as mean ± standard deviation or n (%) unless otherwise specified.
BMI: body mass index; F: female; IVF: *in vitro* fertilisation; M: male.

Table 2. Mean levels of serum markers in preterm and term delivery group[a].

	Preterm delivery ($n=229$)	Term delivery ($n=178$)	p value
Leukocyte ($\times 10^9$/L)	11.8 ± 2.2	10.3 ± 2.1	<.001
Neutrophil ($\times 10^9$/L)	9.4 ± 2.0	7.6 ± 2.1	<.001
Lymphocyte ($\times 10^9$/L)	1.7 ± 5.9	1.8 ± 6.9	.012
Monocyte ($\times 10^9$/L)	0.59 ± 0.3	0.62 ± 0.3	.366
Eosinophil ($\times 10^9$/L)	0.14 ± 0.1	0.15 ± 0.1	.422
Basophil ($\times 10^9$/L)	0.04 ± 0.02	0.05 ± 0.02	.121
Platelet ($\times 10^9$/L)	238.2 ± 69.7	245.6 ± 72.5	.294
NLR	6.8 ± 3.5	4.3 ± 1.6	<.001
PLR	159.6 ± 62.4	140.3 ± 65.3	.003
CRP (mg/dL)	1.79 ± 1.21	1.44 ± 0.84	.001

[a]Data are presented as mean ± standard deviation.
CRP: C-reactive protein; NLR: neutrophil to lymphocyte ratio; PLR: platelet to lymphocyte ratio.

Table 3. Significant predictors of preterm delivery in univariate and multivariate regression analysis.

	Univariate		Multivariate	
Variables	OR (95%CI)	p value	OR (95%CI)	p value
Leukocyte	1.07 (1.01–1.12)	.214	1.01 (0.95–1.07)	.145
Neutrophil	1.10 (0.90–1.24)	.042	1.05 (0.98–1.14)	.095
Lymphocyte	0.91 (0.87–0.95)	.01	0.93 (0.88–0.99)	.212
NLR	1.56 (1.29–1.78)	.001	1.41 (1.32–1.51)	.005
PLR	0.89 (0.79–0.91)	.145	0.92 (0.87–0.98)	.265
CRP	1.15 (1.03–1.29)	.01	1.09 (0.94–1.3)	.324

CI: confidence interval; CRP: C-reactive protein; NLR: neutrophil to lymphocyte ratio; OR: odds ratio; PLR: platelet to lymphocyte ratio.

Discussion

In the present study, we investigated the diagnostic ability of leukocyte subtypes, platelet, CRP, NLR and PLR at admission for predicting delivery time in women with spontaneous, late preterm labour. Compared with patients delivered at term, patients delivered at preterm had significantly higher serum

Table 4. Diagnostic sensitivity and specificity of serum markers in study groups.

	AUC (95%CI)	Sensitivity% (95%CI)	Specificity% (95%CI)	PPV% (95%CI)	NPV% (95%CI)	LR^{+}(95%CI)	LR^{-}(95%CI)	Cut-off value
Leukocyte	0.616 (0.561–0.671)	54.6 (47.9–61.2)	61.2 (53.7–68.4)	64.4 (57.3–71.2)	51.2 (44.2–58.1)	1.41 (1.13–1.75)	0.74 (0.62–0.89)	10.8
Neutrophil	0.623 (0.568–0.678)	52.4 (45.7–59.1)	52.2 (44.6–59.8)	58.5 (51.5–65.4)	46 (39–53.2)	1.1 (0.9–1.34)	0.91 (0.75–1.11)	9.7
Lymphocyte	0.634 (0.580–0.688)	48.9 (42.3–55.6)	51.7 (44.1–59.2)	56.6 (49.3–63.6)	44 (37.2–51)	1.01 (0.83–1.24)	0.99 (0.82–1.20)	1.4
Monocyte	0.508 (0.450–0.566)	34.9 (28.8–41.5)	43.8 (36.4–51.4)	44.4 (37–52)	34.4 (28.2–40.9)	0.62 (0.50–0.77)	1.48 (1.23–1.80)	0.60
Eosinophil	0.519 (0.461–0.576)	39.3 (32.9–45.9)	56.2 (48.6–63.6)	53.6 (45.7–61.3)	41.8 (35.5–48.4)	0.90 (0.71–1.13)	1.08 (0.91–1.28)	0.11
Basophil	0.536 (0.480–0.592)	41.9 (35.4–48.6)	54.5 (46.9–62)	54.2 (46.6–61.7)	42.1 (35.7–48.8)	0.92 (0.74–1.15)	1.07 (0.90–1.27)	0.036
Platelet	0.548 (0.491–0.604)	48 (41.4–54.7)	27 (20.6–34.1)	45.8 (39.4–52.4)	28.7 (22–36.2)	0.66 (0.56–0.77)	1.93 (1.47–2.53)	221
NLR	0.711 (0.662–0.760)	65.1 (58.5–71.2)	62.4 (54.8–69.5)	69 (62.3–75.1)	58.1 (50.8–65.2)	1.73 (1.40–2.14)	0.56 (0.45–0.69)	6.2
PLR	0.590 (0.531–0.649)	45.8 (39.3–52.5)	61.2 (53.7–68.4)	60.3 (52.7–67.7)	46.8 (40.2–53.4)	1.18 (0.94–1.49)	0.88 (0.75–1.04)	147.4
CRP (mg/L)	0.643 (0.588–0.698)	58.1 (51.4–64.5)	59 (51.4–66.3)	64.6 (57.6–71.1)	52.2 (45.1–59.3)	1.42 (1.15–1.74)	0.71 (0.58–0.86)	0.82

AUC: area under curve; CI: confidence interval; CRP: C-reactive protein; LR^{+}: positive likelihood ratio; LR^{-}: negative likelihood ratio; NLR: neutrophil to lymphocyte ratio; PLR: platelet to lymphocyte ratio.

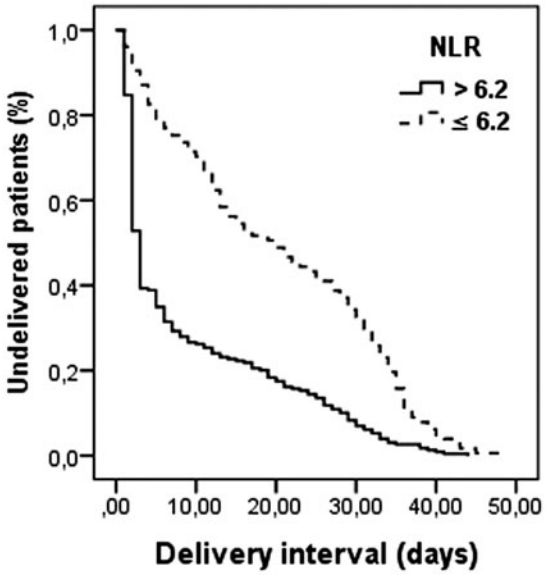

Figure 1. Cox-proportional hazard model of onset of labour to delivery interval according to optimal cut-off point of NLR ($\leq$ 6.2 versus >6.2).

levels of CRP, total leukocyte, neutrophil, NLR and PLR and significantly lower levels of lymphocyte count. Another major finding is that NLR was superior to other markers for discriminating duration of pregnancy.

Prediction and prevention strategies of preterm labour are currently targeted towards the early preterm group, because most of the mortalities occur in babies born at <32 weeks of gestation (Slattery and Morrison 2002). It must be remembered that neonates born between $34^{0/7}$ and $36^{6/7}$ weeks have higher rates of mortality and morbidity compared with those born at term (McParland 2012). As the vast majority of patients who develop preterm labour have no identifiable risk factor, it would be ideal to identify those at low risk for preterm birth earlier and then offer preventative interventions. It is for this reason that women with risk factors for preterm birth (prior history of preterm birth, cervical surgery, uterine anomaly) were excluded from our study population.

Currently, it is evidence-based that the presences of foetal fibronectin in cervicovaginal secretions and short cervix, or a combination of both were recommended in determining women at high risk for preterm birth (American College of Obstetricians and Gynecologists 2001). However, after 34 weeks of gestation, foetal fibronectin begins to break down naturally and can be detected in vaginal secretions (Kiefer and Vintzileos 2008). Similarly, as the cervix physiologically starts to shorten at 32–34 weeks, cervical length measurement is also less useful for predicting preterm birth (Gramellini et al. 2002). Eventually, neither cervical length measurement nor foetal fibronectin test could give enough information regarding the future of pregnancy in patients with preterm labour after 34 weeks of gestation. Therefore, there is a need for the group of spontaneous, late preterm labour to predict the time of delivery interval with efficient markers.

Preterm labour is a syndrome with multiple aetiologies, including intrauterine infection, cervical disease, uterine overdistension, uteroplacental ischaemia or haemorrhage, endocrine disorders, stress and other immunologically-mediated processes (Slattery and Morrison 2002). Although a precise mechanism cannot be established in most cases, a common component of preterm parturition is inflammation at the maternal-foetal interface, which plays a role in initiation and maintenance of labour (Gotsch et al. 2009). Since intrauterine infection is usually chronic and asymptomatic until the membranes rupture, identifying women with those is a major challenge. Although the best-studied site of infection is the amniotic fluid, it is not clear whether amniocentesis improves the outcomes of women with preterm labour. Several markers of intrauterine infection were suggested by previous studies to reduce frequency of preterm delivery (Goldenberg et al. 2000). Interleukin-6, interleukin-8, G-CSF and tumour necrosis factor-alpha from serum samples were introduced as a marker of women with symptoms of preterm labour who later have a preterm delivery but they have not become popular because of their cost and inaccessibility (Goldenberg et al. 2000). Elevated concentrations of inflammation markers such as CRP and leukocyte differential counts in maternal blood have been routinely used in clinical practice to identify the intrauterine infection and subsequent tocolysis failure in patients with preterm labour (Dodds and Iams 1987; Tzur et al. 2013). However, it is demonstrated that the sensitivity and specificity of these markers for detection of positive amniotic fluid culture were of limited diagnostic value (Yoon et al. 1996). Confirming this observation, in the current study, the serum levels of CRP, neutrophil and lymphocyte were significantly differed in univariate analysis but these were not

independently predictive of preterm delivery in multivariate regression analysis.

The design of the present study did not enable correlation between foetal fibronectin and biochemical markers. However, some authors have compared the ability of foetal fibronectin testing with other markers to predict preterm birth in patients with varying degrees of success. Various markers that are associated with preterm birth have been assessed, but very few have shown clinical usefulness compared to fibronectin (Chan 2014). Recent research suggests insulin-like growth factor-binding protein-1 (IGFBP-1) as a potentially promising biomarker. Riboni et al. (2011) and Ting et al. (2007) found that IGFBP-1 was a better marker than foetal fibronectin for predicting preterm delivery. However, little information on the sensitivity, specificity and positive predictive value is known at this time to determine the clinical usefulness of IGFBP-1.

As the physiological immune response to various stressful events is often characterised by an increased number of circulating neutrophils and decreased counts of lymphocytes, some authors proposed to use the NLR as an additional infection marker in clinical practice (Goodman et al. 1995; Zahorec 2001; Wesche et al. 2005; de Jager et al. 2012). Similarly, the ROC curve analysis of our study showed that the AUC for NLR was higher than that of conventional infection markers. In addition, both the PPV and NPV for predicting time of delivery were highest for the NLR. High NLR values have been generally investigated in neoplastic and cardiovascular diseases as an indicator of poor outcome or mortality (Cho et al. 2009; Kaya et al. 2014; Kemal et al. 2014). Moreover, recent data also suggests that NLR is a marker of systemic inflammation and also useful for diagnosis of bacteraemia in infectious emergency admissions. In the past, Goodman et al. (1995) and more recently, de Jager et al. (2012) demonstrated that NLR was better and more sensitive predictor of bacterial infection than routine parameters. In addition, a correlation was also reported between NLR and clinical course severity (Zahorec 2001). Although, to our knowledge, there are no comparative reports, the study by Kim et al. (2011) is similar in some aspects. They compared serum inflammatory markers to predict spontaneous, preterm delivery in preterm labour. Unlike our study, their population was included all preterm labour groups between 22 and 37 weeks of gestation. In their study, foetal fibronectin test was not evaluated and NLR was observed to be the best predictor of preterm delivery after cervical length (Kim et al. 2011). Consequently, we believe that larger, prospective studies are required to validate our findings.

When tissues become inflamed, leucocyte subsets are recruited to the sites of infection by inflammatory cells. Depending on the strength and intensity of the inflammation as well as the resistance and adaptability of the immune system, neutrophilia and lymphocytopenia is developed and maintained (Wesche et al. 2005). The proposed mechanism that elevates NLR is the stimulation of neutrophil stem cells and apoptosis of lymphocytes by a number of hormones and cytokines (Zahorec 2001; Roth and Pircher 2004). In our patients, a NLR >6.2 on admission proved a useful cut-off point for predicting preterm birth, with a sensitivity of 65.1% and a specificity of 62.5%. Similar cut-off values for NLR from patients with infection were represented in some other studies. In the study by Zahorec et al. (2001), normal ratio of neutrophils to lymphocytes was less than 5, whereas under pathologic conditions as severe infection or systemic inflammation, the NLR is increased to higher values than 6.

The limitation of NLR as a marker is that neutrophilia and lymphocytopenia are general innate immune responses to a wide range of stressful events in pregnancy. These factors such as acute infective or inflammatory processes caused by other organs, malnutrition and drug treatments should be assessed as confounding variables and excluded before using NLR as a diagnostic tool for the prediction of preterm birth.

Conclusions

This study explored the association of high NLR with the risk of preterm birth in women with late preterm labour. The importance of NLR is its usefulness in women experiencing spontaneous, preterm labour between 34 and 37 weeks of gestation when the cervical length measurement and foetal fibronectin test are less reliable. NLR is an inexpensive, easily interpretable and promising haematologic parameter that is widely available. The sensitivity and specificity of NLR do not strongly suggest this marker as an effective screening tool in clinical practice. However, concomitant use of markers could be more powerful than measuring any of the individual markers alone. We therefore suggest that NLR could be used in combination with existing (cervical length measurement and/or foetal fibronectin test) or new markers to improve detection rates of preterm birth. Further prospective researches combining NLR with other predictors are needed to verify findings demonstrated in this study.

Disclosure statement

The authors have no conflicts of interest relevant to this article.

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